PGCP-AC AUGUST 26

Database Assignment 1

Schema Design & Table Creation (DDL & DML Commands)

1. Define a schema for a Library Management System with the following entities:
- **Books**
- **Authors**
- **Members**
- **Borrow_Records**

2. Write the SQL command to create a table Authors with the following fields:
- **author_id (Primary Key, INT)**
- **name (VARCHAR(100))**
- **country (VARCHAR(50))**

3. Write the SQL command to create a table Books with the following fields:
- **book_id (Primary Key, INT)**
- **title (VARCHAR(150))**
- **author_id (Foreign Key referencing Authors)**
- **published_year (YEAR)**
- **available_copies (INT)**

4. Write the SQL command to create a table Members with:
- **member_id (Primary Key, INT)**
- **name (VARCHAR(100))**
- **email (VARCHAR(100), unique)**
- **phone (VARCHAR(15))**

5. Write the SQL command to create a table Borrow_Records with:
- **record_id (Primary Key, INT)**
- **member_id (Foreign Key referencing Members)**
- **book_id (Foreign Key referencing Books)**
- **borrow_date (DATE)**
- **return_date (DATE)**

6. Modify the Books table to add a column genre of type VARCHAR(50).

7. Write the SQL command to drop the Borrow_Records table.

8. Insert 3 records into the Authors table.

9. Insert 5 books into the Books table.

10. Insert 3 members into the Members table.

11. Insert 4 borrow records into the Borrow_Records table.

12. Write an SQL query to select all books where published_year is after 2015.

13. Write a SQL query to create a foreign key & primary key relationship between two tables.

14. Write an SQL query to find all members who have borrowed the book with title 'Database Systems'.

15. Update the available_copies column of a specific book (choose any book) by reducing it by 1 after it is borrowed.

16. Delete a record from Members where member_id = 3.

17. Update a Book name record from Book table with id = 1.

18. Write an SQL query to list all books along with their authors' names.

19. Write an SQL query to delete all books from the Books table where the published_year is before 2000.

20. Write an SQL query to find all books that are never borrowed (i.e., no records in Borrow_Records).

```
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| cdac               |
| day3               |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
6 rows in set (0.15 sec)

mysql> CREATE DATABASE library_management_system;
Query OK, 1 row affected (0.02 sec)

mysql> USE library_management_system;
Database changed
mysql> CREATE TABLE Author (author_id INT PRIMARY KEY, name VARCHAR(100), country VARCHAR(50));
Query OK, 0 rows affected (0.09 sec)

mysql> CREATE TABLE Books (
    ->     book_id INT PRIMARY KEY,
    ->     title VARCHAR(150),
    ->     author_id INT REFERENCES Authors(author_id),
    ->     published_year YEAR,
    ->     available_copies INT
```

```
   -> );
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE Members (member_id INT PRIMARY KEY, name  VARCHAR(100), email VARCHAR(100) UNIQUE,
phone VARCHAR(15));
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE Borrow_Records (
   -> record_id INT PRIMARY KEY,
   -> member_id REFERENCES Book(book_id),
   -> borrow_date DATE,
   -> return_date DATE^C
mysql> CREATE TABLE Borrow_Records (
   ->     record_id INT PRIMARY KEY,
   ->     member_id INT REFERENCES Members(member_id),
   ->     book_id INT REFERENCES Books(book_id),
   ->     borrow_date DATE,
   ->     return_date DATE
   -> );
Query OK, 0 rows affected (0.03 sec)

mysql> desc Borrow_Records;
+-------------+------+------+-----+---------+-------+
| Field       | Type | Null | Key | Default | Extra |
+-------------+------+------+-----+---------+-------+
| record_id   | int  | NO   | PRI | NULL    |       |
| member_id   | int  | YES  |     | NULL    |       |
| book_id     | int  | YES  |     | NULL    |       |
| borrow_date | date | YES  |     | NULL    |       |
| return_date | date | YES  |     | NULL    |       |
+-------------+------+------+-----+---------+-------+
5 rows in set (0.11 sec)

mysql> show tables;
```

```
+------------------------------------+
| Tables_in_library_management_system |
+------------------------------------+
| author                             |
| books                              |
| borrow_records                     |
| members                            |
+------------------------------------+
4 rows in set (0.04 sec)

mysql> desc author;
+-----------+--------------+------+-----+---------+-------+
| Field     | Type         | Null | Key | Default | Extra |
+-----------+--------------+------+-----+---------+-------+
| author_id | int          | NO   | PRI | NULL    |       |
| name      | varchar(100) | YES  |     | NULL    |       |
| country   | varchar(50)  | YES  |     | NULL    |       |
+-----------+--------------+------+-----+---------+-------+
3 rows in set (0.00 sec)

mysql> desc books;
+------------------+--------------+------+-----+---------+-------+
| Field            | Type         | Null | Key | Default | Extra |
+------------------+--------------+------+-----+---------+-------+
| book_id          | int          | NO   | PRI | NULL    |       |
| title            | varchar(150) | YES  |     | NULL    |       |
| author_id        | int          | YES  |     | NULL    |       |
| published_year   | year         | YES  |     | NULL    |       |
| available_copies | int          | YES  |     | NULL    |       |
+------------------+--------------+------+-----+---------+-------+
5 rows in set (0.00 sec)

mysql> desc borrow_records;
+-------------+------+------+-----+---------+-------+
```

| Field | Type | Null | Key | Default | Extra |

+-------------+------+------+-----+---------+-------+

| record_id | int | NO | PRI | NULL | |

| member_id | int | YES | | NULL | |

| book_id | int | YES | | NULL | |

| borrow_date | date | YES | | NULL | |

| return_date | date | YES | | NULL | |

+-------------+------+------+-----+---------+-------+

5 rows in set (0.00 sec)

mysql> desc members;

+-----------+--------------+------+-----+---------+-------+

| Field | Type | Null | Key | Default | Extra |

+-----------+--------------+------+-----+---------+-------+

| member_id | int | NO | PRI | NULL | |

| name | varchar(100) | YES | | NULL | |

| email | varchar(100) | YES | UNI | NULL | |

| phone | varchar(15) | YES | | NULL | |

+-----------+--------------+------+-----+---------+-------+

4 rows in set (0.00 sec)

mysql> ALTER TABLE Books ADD genre VARCHAR(50);

Query OK, 0 rows affected (0.22 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> DROP TABLE Borrow_Records;

Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO author VALUES ((1, 'Nilesh', 'Pune'), (2, 'Yash', 'Malegoan'), (3, 'Gayatri', 'Chakan'));

ERROR 1241 (21000): Operand should contain 1 column(s)

mysql> INSERT INTO author VALUES {(1, 'Nilesh', 'Pune'), (2, 'Yash', 'Malegoan'), (3, 'Gayatri', 'Chakan')};

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '{(1, 'Nilesh', 'Pune'), (2, 'Yash', 'Malegoan'), (3, 'Gayatri', 'Chakan')}' at line 1

mysql> INSERT INTO author VALUES (1, 'Nilesh', 'Pune');

```
Query OK, 1 row affected (0.03 sec)

mysql> INSERT INTO author VALUES (3, 'Gayatri', 'Chakan');
Query OK, 1 row affected (0.02 sec)

mysql> INSERT INTO author VALUES (2, 'Yash', 'Malegoan');
Query OK, 1 row affected (0.03 sec)

mysql> SELECT * FROM AUTHOR;
+-----------+---------+----------+
| author_id | name    | country  |
+-----------+---------+----------+
|         1 | Nilesh  | Pune     |
|         2 | Yash    | Malegoan |
|         3 | Gayatri | Chakan   |
+-----------+---------+----------+
3 rows in set (0.02 sec)

mysql>  INSERT INTO author VALUES (1, 'R. Sharma', 'India');
ERROR 1062 (23000): Duplicate entry '1' for key 'author.PRIMARY'
mysql>
mysql>  INSERT INTO author VALUES (2, 'A. Verma', 'USA');
ERROR 1062 (23000): Duplicate entry '2' for key 'author.PRIMARY'
mysql>
mysql>  INSERT INTO author VALUES (3, 'P. Kumar', 'India');
ERROR 1062 (23000): Duplicate entry '3' for key 'author.PRIMARY'
mysql> TRUNCATE TABLE author;
Query OK, 0 rows affected (0.06 sec)

mysql>  INSERT INTO author VALUES (1, 'R. Sharma', 'India');
Query OK, 1 row affected (0.03 sec)

mysql> INSERT INTO author VALUES (2, 'Yash', 'Malegoan');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO author VALUES (3, 'Gayatri', 'Chakan');
Query OK, 1 row affected (0.02 sec)

mysql> SELECT * FROM AUTHOR;
+-----------+-----------+----------+
| author_id | name      | country  |
+-----------+-----------+----------+
|         1 | R. Sharma | India    |
|         2 | Yash      | Malegoan |
|         3 | Gayatri   | Chakan   |
+-----------+-----------+----------+
3 rows in set (0.00 sec)

mysql> INSERT INTO books VALUES
    -> (101, 'SQL Basics', 1, 2018, 5),
    -> (102, 'Database Systems', 2, 2020, 3),
    -> (103, 'C Programming', 3, 2015, 4),3
    -> (101, 'Java Fundamentals', 1, 2022, 6),
    -> (105, 'Web Development', 2, 2019, 2);^C
mysql> INSERT INTO Books VALUES
    ->     (101, 'SQL Basics', 1, 2018, 5),
    ->     (102, 'Database Systems', 2, 2020, 3),
    ->     (103, 'C Programming', 3, 2015, 4),
    ->     (104, 'Java Fundamentals', 1, 2022, 6),
    ->     (105, 'Web Development', 2, 2019, 2);
ERROR 1136 (21S01): Column count doesn't match value count at row 1
mysql> desc books;
+------------------+--------------+------+-----+---------+-------+
| Field            | Type         | Null | Key | Default | Extra |
+------------------+--------------+------+-----+---------+-------+
| book_id          | int          | NO   | PRI | NULL    |       |
| title            | varchar(150) | YES  |     | NULL    |       |
| author_id        | int          | YES  |     | NULL    |       |
```

```
| published_year   | year        | YES |     | NULL    |       |
| available_copies | int         | YES |     | NULL    |       |
| genre            | varchar(50) | YES |     | NULL    |       |
+------------------+-------------+-----+-----+---------+-------+
6 rows in set (0.00 sec)

mysql> Alter table books remove genre;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'genre' at line 1
mysql> Alter table books drop column genre;
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc books;
+------------------+-------------+------+-----+---------+-------+
| Field            | Type        | Null | Key | Default | Extra |
+------------------+-------------+------+-----+---------+-------+
| book_id          | int         | NO   | PRI | NULL    |       |
| title            | varchar(150) | YES |     | NULL    |       |
| author_id        | int         | YES |     | NULL    |       |
| published_year   | year        | YES |     | NULL    |       |
| available_copies | int         | YES |     | NULL    |       |
+------------------+-------------+------+-----+---------+-------+
5 rows in set (0.00 sec)

mysql> INSERT INTO Books VALUES
    ->     (101, 'SQL Basics', 1, 2018, 5),
    ->     (102, 'Database Systems', 2, 2020, 3),
    ->     (103, 'C Programming', 3, 2015, 4),
    ->     (104, 'Java Fundamentals', 1, 2022, 6),
    ->     (105, 'Web Development', 2, 2019, 2);
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO members VALUES
    -> (1, 'Amit Patil', 'amit@example', '9876543210'),
    -> ( 2, 'Priya Shah', 'priya@example.com', '9876543211'),
    -> (3, 'Rahul Joshi', 'rahul@example.com', '9876543212');
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> show tables;
+-----------------------------------+
| Tables_in_library_management_system |
+-----------------------------------+
| author                            |
| books                             |
| members                           |
+-----------------------------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM author;
+-----------+-----------+----------+
| author_id | name      | country  |
+-----------+-----------+----------+
|         1 | R. Sharma | India    |
|         2 | Yash      | Malegoan |
|         3 | Gayatri   | Chakan   |
+-----------+-----------+----------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM books;
+---------+------------------+-----------+----------------+------------------+
| book_id | title            | author_id | published_year | available_copies |
+---------+------------------+-----------+----------------+------------------+
|     101 | SQL Basics       |         1 |           2018 |                5 |
|     102 | Database Systems |         2 |           2020 |                3 |
|     103 | C Programming    |         3 |           2015 |                4 |
```

```
|     104 | Java Fundamentals |        1 |         2022 |             6 |
|     105 | Web Development   |        2 |         2019 |             2 |
+---------+-------------------+----------+--------------+-----------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM memebers;
ERROR 1146 (42S02): Table 'library_management_system.memebers' doesn't exist
mysql> SELECT * FROM members;
+-----------+-------------+-------------------+------------+
| member_id | name        | email             | phone      |
+-----------+-------------+-------------------+------------+
|         1 | Amit Patil  | amit@example      | 9876543210 |
|         2 | Priya Shah  | priya@example.com | 9876543211 |
|         3 | Rahul Joshi | rahul@example.com | 9876543212 |
+-----------+-------------+-------------------+------------+
3 rows in set (0.00 sec)

mysql> desc borrow_records;
ERROR 1146 (42S02): Table 'library_management_system.borrow_records' doesn't exist
mysql> show tables;
+-------------------------------------+
| Tables_in_library_management_system |
+-------------------------------------+
| author                              |
| books                               |
| members                             |
+-------------------------------------+
3 rows in set (0.00 sec)

mysql> CREATE TABLE borrow_records (record_id int primary key, member_id int references Members(member_id),
book_id int references Books(book_id), borrow_date DATE, return_date DATE);
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO  borrow_records VALUES
```

```
    -> (1, 1, 101, '2026-10-01', '2026-10-10'),
    -> (2, 2, 102, '2026-10-02', '2026-10-12'),
    -> (3, 3, 103, '2026-10-03', NULL),
    -> (4, 1, 104, '2026-10-05', NULL);
Query OK, 4 rows affected (0.03 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM borrow_table;
ERROR 1146 (42S02): Table 'library_management_system.borrow_table' doesn't exist
mysql> SELECT * FROM borrow_records;
+-----------+-----------+---------+-------------+-------------+
| record_id | member_id | book_id | borrow_date | return_date |
+-----------+-----------+---------+-------------+-------------+
|         1 |         1 |     101 | 2026-10-01  | 2026-10-10  |
|         2 |         2 |     102 | 2026-10-02  | 2026-10-12  |
|         3 |         3 |     103 | 2026-10-03  | NULL        |
|         4 |         1 |     104 | 2026-10-05  | NULL        |
+-----------+-----------+---------+-------------+-------------+
4 rows in set (0.00 sec)

mysql> SELECT * FROM books WHERE published_year > 2015;
+---------+-------------------+-----------+----------------+------------------+
| book_id | title             | author_id | published_year | available_copies |
+---------+-------------------+-----------+----------------+------------------+
|     101 | SQL Basics        |         1 |           2018 |                5 |
|     102 | Database Systems  |         2 |           2020 |                3 |
|     104 | Java Fundamentals |         1 |           2022 |                6 |
|     105 | Web Development   |         2 |           2019 |                2 |
+---------+-------------------+-----------+----------------+------------------+
4 rows in set (0.03 sec)

mysql> CREATE TABLE Departments (department_id INT PRIMARY KEY, department_name VARCHAR(100));
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> CREATE TABLE Students (student_id INT PRIMARY KEY, department_id REFERENCES
Departments(department_id));

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server
version for the right syntax to use near 'REFERENCES Departments(department_id))' at line 1

mysql> CREATE TABLE Students

   -> ( student_id INT PRIMARY KEY,

   -> department_id REFERENCES Departments(department_id)

   -> );

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server
version for the right syntax to use near 'REFERENCES Departments(department_id)

)' at line 3

mysql> CREATE TABLE Students (

   -> student_id INT PRIMARY KEY,

   -> department_id INT REFERENCES Departments(department_id)

   -> );

Query OK, 0 rows affected (0.04 sec)


mysql> SELECT * FROM Memebers JOIN Borrow_Records JOIN Books;

ERROR 1146 (42S02): Table 'library_management_system.memebers' doesn't exist

mysql> show tables;

+-------------------------------------+

| Tables_in_library_management_system |

+-------------------------------------+

| author                              |

| books                               |

| borrow_records                      |

| departments                         |

| members                             |

| students                            |

+-------------------------------------+

6 rows in set (0.00 sec)


mysql> SELECT * FROM Members JOIN Borrow_Records JOIN Books;

+-----------+-------------+-------------------+------------+-----------+-----------+---------+-------------+-------------+---------+-----------------
---+-----------+---------------+-----------------+
```

| member_id | name | email | phone | record_id | member_id | book_id | borrow_date | return_date | book_id | title | author_id | published_year | available_copies |

+-----------+-------------+-------------------+------------+-----------+-----------+---------+-------------+-------------+---------+----------------+-----------+---------------+-----------------+

| 1 | Amit Patil | amit@example | 9876543210 | 4 | 1 | 104 | 2026-10-05 | NULL | 101 | SQL Basics | 1 | 2018 | 5 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 4 | 1 | 104 | 2026-10-05 | NULL | 101 | SQL Basics | 1 | 2018 | 5 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 4 | 1 | 104 | 2026-10-05 | NULL | 101 | SQL Basics | 1 | 2018 | 5 |

| 1 | Amit Patil | amit@example | 9876543210 | 3 | 3 | 103 | 2026-10-03 | NULL | 101 | SQL Basics | 1 | 2018 | 5 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 3 | 3 | 103 | 2026-10-03 | NULL | 101 | SQL Basics | 1 | 2018 | 5 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 3 | 3 | 103 | 2026-10-03 | NULL | 101 | SQL Basics | 1 | 2018 | 5 |

| 1 | Amit Patil | amit@example | 9876543210 | 2 | 2 | 102 | 2026-10-02 | 2026-10-12 | 101 | SQL Basics | 1 | 2018 | 5 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 2 | 2 | 102 | 2026-10-02 | 2026-10-12 | 101 | SQL Basics | 1 | 2018 | 5 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 2 | 2 | 102 | 2026-10-02 | 2026-10-12 | 101 | SQL Basics | 1 | 2018 | 5 |

| 1 | Amit Patil | amit@example | 9876543210 | 1 | 1 | 101 | 2026-10-01 | 2026-10-10 | 101 | SQL Basics | 1 | 2018 | 5 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 1 | 1 | 101 | 2026-10-01 | 2026-10-10 | 101 | SQL Basics | 1 | 2018 | 5 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 1 | 1 | 101 | 2026-10-01 | 2026-10-10 | 101 | SQL Basics | 1 | 2018 | 5 |

| 1 | Amit Patil | amit@example | 9876543210 | 4 | 1 | 104 | 2026-10-05 | NULL | 102 | Database Systems | 2 | 2020 | 3 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 4 | 1 | 104 | 2026-10-05 | NULL | 102 | Database Systems | 2 | 2020 | 3 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 4 | 1 | 104 | 2026-10-05 | NULL | 102 | Database Systems | 2 | 2020 | 3 |

| 1 | Amit Patil | amit@example | 9876543210 | 3 | 3 | 103 | 2026-10-03 | NULL | 102 | Database Systems | 2 | 2020 | 3 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 3 | 3 | 103 | 2026-10-03 | NULL | 102 | Database Systems | 2 | 2020 | 3 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 3 | 3 | 103 | 2026-10-03 | NULL | 102 | Database Systems | 2 | 2020 | 3 |

| 1 | Amit Patil | amit@example | 9876543210 | 2 | 2 | 102 | 2026-10-02 | 2026-10-12 | 102 | Database Systems | 2 | 2020 | 3 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 2 | 2 | 102 | 2026-10-02 | 2026-10-12 | 102 | Database Systems | 2 | 2020 | 3 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 2 | 2 | 102 | 2026-10-02 | 2026-10-12 | 102 | Database Systems | 2 | 2020 | 3 |

| 1 | Amit Patil | amit@example | 9876543210 | 1 | 1 | 101 | 2026-10-01 | 2026-10-10 | 102 | Database Systems | 2 | 2020 | 3 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 1 | 1 | 101 | 2026-10-01 | 2026-10-10 | 102 | Database Systems | 2 | 2020 | 3 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 1 | 1 | 101 | 2026-10-01 | 2026-10-10 | 102 | Database Systems | 2 | 2020 | 3 |

| 1 | Amit Patil | amit@example | 9876543210 | 4 | 1 | 104 | 2026-10-05 | NULL | 103 | C Programming | 3 | 2015 | 4 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 4 | 1 | 104 | 2026-10-05 | NULL | 103 | C Programming | 3 | 2015 | 4 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 4 | 1 | 104 | 2026-10-05 | NULL | 103 | C Programming | 3 | 2015 | 4 |

| 1 | Amit Patil | amit@example | 9876543210 | 3 | 3 | 103 | 2026-10-03 | NULL | 103 | C Programming | 3 | 2015 | 4 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 3 | 3 | 103 | 2026-10-03 | NULL | 103 | C Programming | 3 | 2015 | 4 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 3 | 3 | 103 | 2026-10-03 | NULL | 103 | C Programming | 3 | 2015 | 4 |

| 1 | Amit Patil | amit@example | 9876543210 | 2 | 2 | 102 | 2026-10-02 | 2026-10-12 | 103 | C Programming | 3 | 2015 | 4 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 2 | 2 | 102 | 2026-10-02 | 2026-10-12 | 103 | C Programming | 3 | 2015 | 4 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 2 | 2 | 102 | 2026-10-02 | 2026-10-12 | 103 | C Programming | 3 | 2015 | 4 |

| 1 | Amit Patil | amit@example | 9876543210 | 1 | 1 | 101 | 2026-10-01 | 2026-10-10 | 103 | C Programming | 3 | 2015 | 4 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 1 | 1 | 101 | 2026-10-01 | 2026-10-10 | 103 | C Programming | 3 | 2015 | 4 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 1 | 1 | 101 | 2026-10-01 | 2026-10-10 | 103 | C Programming | 3 | 2015 | 4 |

| 1 | Amit Patil | amit@example | 9876543210 | 4 | 1 | 104 | 2026-10-05 | NULL | 104 | Java Fundamentals | 1 | 2022 | 6 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 4 | 1 | 104 | 2026-10-05 | NULL | 104 | Java Fundamentals | 1 | 2022 | 6 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 4 | 1 | 104 | 2026-10-05 | NULL | 104 | Java Fundamentals | 1 | 2022 | 6 |

| 1 | Amit Patil | amit@example | 9876543210 | 3 | 3 | 103 | 2026-10-03 | NULL | 104 | Java Fundamentals | 1 | 2022 | 6 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 3 | 3 | 103 | 2026-10-03 | NULL | 104 | Java Fundamentals | 1 | 2022 | 6 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 3 | 3 | 103 | 2026-10-03 | NULL | 104 | Java Fundamentals | 1 | 2022 | 6 |

| 1 | Amit Patil | amit@example | 9876543210 | 2 | 2 | 102 | 2026-10-02 | 2026-10-12 | 104 | Java Fundamentals | 1 | 2022 | 6 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 2 | 2 | 102 | 2026-10-02 | 2026-10-12 | 104 | Java Fundamentals | 1 | 2022 | 6 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 2 | 2 | 102 | 2026-10-02 | 2026-10-12 | 104 | Java Fundamentals | 1 | 2022 | 6 |

| 1 | Amit Patil | amit@example | 9876543210 | 1 | 1 | 101 | 2026-10-01 | 2026-10-10 | 104 | Java Fundamentals | 1 | 2022 | 6 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 1 | 1 | 101 | 2026-10-01 | 2026-10-10 | 104 | Java Fundamentals | 1 | 2022 | 6 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 1 | 1 | 101 | 2026-10-01 | 2026-10-10 | 104 | Java Fundamentals | 1 | 2022 | 6 |

| 1 | Amit Patil | amit@example | 9876543210 | 4 | 1 | 104 | 2026-10-05 | NULL | 105 | Web Development | 2 | 2019 | 2 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 4 | 1 | 104 | 2026-10-05 | NULL | 105 | Web Development | 2 | 2019 | 2 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 4 | 1 | 104 | 2026-10-05 | NULL | 105 | Web Development | 2 | 2019 | 2 |

| 1 | Amit Patil | amit@example | 9876543210 | 3 | 3 | 103 | 2026-10-03 | NULL | 105 | Web Development | 2 | 2019 | 2 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 3 | 3 | 103 | 2026-10-03 | NULL | 105 | Web Development | 2 | 2019 | 2 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 3 | 3 | 103 | 2026-10-03 | NULL | 105 | Web Development | 2 | 2019 | 2 |

| 1 | Amit Patil | amit@example | 9876543210 | 2 | 2 | 102 | 2026-10-02 | 2026-10-12 | 105 | Web Development | 2 | 2019 | 2 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 2 | 2 | 102 | 2026-10-02 | 2026-10-12 | 105 | Web Development | 2 | 2019 | 2 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 2 | 2 | 102 | 2026-10-02 | 2026-10-12 | 105 | Web Development | 2 | 2019 | 2 |

| 1 | Amit Patil | amit@example | 9876543210 | 1 | 1 | 101 | 2026-10-01 | 2026-10-10 | 105 | Web Development | 2 | 2019 | 2 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 1 | 1 | 101 | 2026-10-01 | 2026-10-10 | 105 | Web Development | 2 | 2019 | 2 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 1 | 1 | 101 | 2026-10-01 | 2026-10-10 | 105 | Web Development | 2 | 2019 | 2 |

+----------+-------------+-------------------+------------+-----------+-----------+---------+-------------+-------------+---------+-----------------+-----------+----------------+-----------------+

60 rows in set (0.00 sec)

SELECT * FROM Members JOIN Borrow_Records JOIN Books WHERE book_title = 'Database Systems';

ERROR 1054 (42S22): Unknown column 'book_title' in 'where clause'

mysql> SELECT * FROM Members JOIN Borrow_Records JOIN Books WHERE book_title='Database Systems';

ERROR 1054 (42S22): Unknown column 'book_title' in 'where clause'

mysql> desc books;

+-----------------+--------------+------+-----+---------+-------+

| Field | Type | Null | Key | Default | Extra |

+-----------------+--------------+------+-----+---------+-------+

| book_id | int | NO | PRI | NULL | |

| title | varchar(150) | YES | | NULL | |

| author_id | int | YES | | NULL | |

| published_year | year | YES | | NULL | |

| available_copies | int | YES | | NULL | |

+-----------------+--------------+------+-----+---------+-------+

5 rows in set (0.00 sec)

mysql> SELECT * FROM Members JOIN Borrow_Records JOIN Books WHERE title='Database Systems';

+-----------+-------------+------------------+------------+-----------+-----------+---------+-------------+-------------+---------+----------------+-----------+---------------+-----------------+

| member_id | name | email | phone | record_id | member_id | book_id | borrow_date | return_date | book_id | title | author_id | published_year | available_copies |

+-----------+-------------+------------------+------------+-----------+-----------+---------+-------------+-------------+---------+----------------+-----------+---------------+-----------------+

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 1 | 1 | 101 | 2026-10-01 | 2026-10-10 | 102 | Database Systems | 2 | 2020 | 3 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 1 | 1 | 101 | 2026-10-01 | 2026-10-10 | 102 | Database Systems | 2 | 2020 | 3 |

| 1 | Amit Patil | amit@example | 9876543210 | 1 | 1 | 101 | 2026-10-01 | 2026-10-10 | 102 | Database Systems | 2 | 2020 | 3 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 2 | 2 | 102 | 2026-10-02 | 2026-10-12 | 102 | Database Systems | 2 | 2020 | 3 |

| 2 | Priya Shah | priya@example.com | 9876543211 | 2 | 2 | 102 | 2026-10-02 | 2026-10-12 | 102 | Database Systems | 2 | 2020 | 3 |

| 1 | Amit Patil | amit@example | 9876543210 | 2 | 2 | 102 | 2026-10-02 | 2026-10-12 | 102 | Database Systems | 2 | 2020 | 3 |

| 3 | Rahul Joshi | rahul@example.com | 9876543212 | 3 | 3 | 103 | 2026-10-03 | NULL | 102 | Database Systems | 2 | 2020 | 3 |

```
|         2 | Priya Shah  | priya@example.com | 9876543211 |         3 |         3 |     103 | 2026-10-03  | NULL       |     102 | Database Systems |         2 |          2020 |             3 |
|         1 | Amit Patil  | amit@example      | 9876543210 |         3 |         3 |     103 | 2026-10-03  | NULL       |     102 | Database Systems |         2 |          2020 |             3 |
|         3 | Rahul Joshi | rahul@example.com | 9876543212 |         4 |         1 |     104 | 2026-10-05  | NULL       |     102 | Database Systems |         2 |          2020 |             3 |
|         2 | Priya Shah  | priya@example.com | 9876543211 |         4 |         1 |     104 | 2026-10-05  | NULL       |     102 | Database Systems |         2 |          2020 |             3 |
|         1 | Amit Patil  | amit@example      | 9876543210 |         4 |         1 |     104 | 2026-10-05  | NULL       |     102 | Database Systems |         2 |          2020 |             3 |
+-----------+-------------+-------------------+------------+-----------+-----------+---------+-------------+-------------+---------+------------------+-----------+---------------+-----------------+
12 rows in set (0.00 sec)

mysql> SELECT Members.name
    -> FROM Members
    -> JOIN Borrow_Records
    ->     ON Members.member_id = Borrow_Records.member_id
    -> JOIN Books
    ->     ON Borrow_Records.book_id = Books.book_id
    -> WHERE Books.title = 'Database Systems';
+-------------+
| name        |
+-------------+
| Priya Shah  |
+-------------+
1 row in set (0.00 sec)

mysql> SELECT *
    -> FROM Members
    -> JOIN Borrow_Records
    ->     ON Members.member_id = Borrow_Records.member_id
    -> JOIN Books
    ->     ON Borrow_Records.book_id = Books.book_id
    -> WHERE Books.title = 'Database Systems';
+-----------+-------------+-------------------+------------+-----------+-----------+---------+-------------+-------------+---------+------------------+-----------+---------------+-----------------+
```

| member_id | name | email | phone | record_id | member_id | book_id | borrow_date | return_date | book_id | title | author_id | published_year | available_copies |

+-----------+------------+------------------+------------+-----------+-----------+---------+-------------+-------------+---------+----------------+-----------+----------------+------------------+

| 2 | Priya Shah | priya@example.com | 9876543211 | 2 | 2 | 102 | 2026-10-02 | 2026-10-12 | 102 | Database Systems | 2 | 2020 | 3 |

+-----------+------------+------------------+------------+-----------+-----------+---------+-------------+-------------+---------+----------------+-----------+----------------+------------------+

1 row in set (0.00 sec)

mysql> desc books;

+------------------+--------------+------+-----+---------+-------+

| Field | Type | Null | Key | Default | Extra |

+------------------+--------------+------+-----+---------+-------+

| book_id | int | NO | PRI | NULL | |

| title | varchar(150) | YES | | NULL | |

| author_id | int | YES | | NULL | |

| published_year | year | YES | | NULL | |

| available_copies | int | YES | | NULL | |

+------------------+--------------+------+-----+---------+-------+

5 rows in set (0.00 sec)

mysql> SELECT * FROM books;

+---------+------------------+-----------+----------------+------------------+

| book_id | title | author_id | published_year | available_copies |

+---------+------------------+-----------+----------------+------------------+

| 101 | SQL Basics | 1 | 2018 | 5 |

| 102 | Database Systems | 2 | 2020 | 3 |

| 103 | C Programming | 3 | 2015 | 4 |

| 104 | Java Fundamentals | 1 | 2022 | 6 |

| 105 | Web Development | 2 | 2019 | 2 |

+---------+------------------+-----------+----------------+------------------+

5 rows in set (0.00 sec)

mysql> UPDATE TABLE books SET available_copies=available_copies-1 WHERE books_id=103;

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'TABLE books SET available_copies=available_copies-1 WHERE books_id=103' at line 1

mysql> UPDATE TABLE books SET available_copies=available_copies-1 WHERE book_id=103;

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'TABLE books SET available_copies=available_copies-1 WHERE book_id=103' at line 1

mysql> UPDATE books SET available_copies=available_copies-1 WHERE book_id=103;

Query OK, 1 row affected (0.07 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM books;

+---------+------------------+-----------+----------------+------------------+

| book_id | title | author_id | published_year | available_copies |

+---------+------------------+-----------+----------------+------------------+

| 101 | SQL Basics | 1 | 2018 | 5 |

| 102 | Database Systems | 2 | 2020 | 3 |

| 103 | C Programming | 3 | 2015 | 3 |

| 104 | Java Fundamentals | 1 | 2022 | 6 |

| 105 | Web Development | 2 | 2019 | 2 |

+---------+------------------+-----------+----------------+------------------+

5 rows in set (0.00 sec)

mysql> DELETE FROM members where member_id=3;

Query OK, 1 row affected (0.03 sec)

mysql> SELECT * FROM members;

+-----------+------------+-------------------+------------+

| member_id | name | email | phone |

+-----------+------------+-------------------+------------+

| 1 | Amit Patil | amit@example | 9876543210 |

| 2 | Priya Shah | priya@example.com | 9876543211 |

+-----------+------------+-------------------+------------+

2 rows in set (0.00 sec)

mysql> SELECT * FROM books;

```
+---------+------------------+-----------+----------------+------------------+
| book_id | title            | author_id | published_year | available_copies |
+---------+------------------+-----------+----------------+------------------+
|     101 | SQL Basics       |         1 |           2018 |                5 |
|     102 | Database Systems |         2 |           2020 |                3 |
|     103 | C Programming    |         3 |           2015 |                3 |
|     104 | Java Fundamentals |         1 |           2022 |                6 |
|     105 | Web Development  |         2 |           2019 |                2 |
+---------+------------------+-----------+----------------+------------------+
5 rows in set (0.00 sec)

mysql> UPDATE books SET title='Advanced SQL' WHERE book_id=1;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql> SELECT * FROM books;
+---------+------------------+-----------+----------------+------------------+
| book_id | title            | author_id | published_year | available_copies |
+---------+------------------+-----------+----------------+------------------+
|     101 | SQL Basics       |         1 |           2018 |                5 |
|     102 | Database Systems |         2 |           2020 |                3 |
|     103 | C Programming    |         3 |           2015 |                3 |
|     104 | Java Fundamentals |         1 |           2022 |                6 |
|     105 | Web Development  |         2 |           2019 |                2 |
+---------+------------------+-----------+----------------+------------------+
5 rows in set (0.00 sec)

mysql> UPDATE books SET title='Advanced SQL' WHERE book_id=101;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM books;
+---------+------------------+-----------+----------------+------------------+
| book_id | title            | author_id | published_year | available_copies |
```

```
+---------+------------------+-----------+----------------+------------------+
|     101 | Advanced SQL     |         1 |           2018 |                5 |
|     102 | Database Systems |         2 |           2020 |                3 |
|     103 | C Programming    |         3 |           2015 |                3 |
|     104 | Java Fundamentals |         1 |           2022 |                6 |
|     105 | Web Development  |         2 |           2019 |                2 |
+---------+------------------+-----------+----------------+------------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM books;
+---------+------------------+-----------+----------------+------------------+
| book_id | title            | author_id | published_year | available_copies |
+---------+------------------+-----------+----------------+------------------+
|     101 | Advanced SQL     |         1 |           2018 |                5 |
|     102 | Database Systems |         2 |           2020 |                3 |
|     103 | C Programming    |         3 |           2015 |                3 |
|     104 | Java Fundamentals |         1 |           2022 |                6 |
|     105 | Web Development  |         2 |           2019 |                2 |
+---------+------------------+-----------+----------------+------------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM authors;
ERROR 1146 (42S02): Table 'library_management_system.authors' doesn't exist
mysql> SELECT * FROM author;
+-----------+-----------+----------+
| author_id | name      | country  |
+-----------+-----------+----------+
|         1 | R. Sharma | India    |
|         2 | Yash      | Malegoan |
|         3 | Gayatri   | Chakan   |
+-----------+-----------+----------+
3 rows in set (0.00 sec)

mysql> SELECT title from books JOIN author ON books.author_id-author.author_id;
```

```
+-------------------+
| title             |
+-------------------+
| Advanced SQL      |
| Advanced SQL      |
| Database Systems  |
| Database Systems  |
| C Programming     |
| C Programming     |
| Java Fundamentals |
| Java Fundamentals |
| Web Development   |
| Web Development   |
+-------------------+
10 rows in set (0.00 sec)

mysql> SELECT title from books JOIN author ON books.author_id=author.author_id;
+-------------------+
| title             |
+-------------------+
| Advanced SQL      |
| Database Systems  |
| C Programming     |
| Java Fundamentals |
| Web Development   |
+-------------------+
5 rows in set (0.00 sec)

mysql> SELECT title, name from books JOIN author ON books.author_id=author.author_id;
+-------------------+-----------+
| title             | name      |
+-------------------+-----------+
| Advanced SQL      | R. Sharma |
| Database Systems  | Yash      |
```

| C Programming | Gayatri |

| Java Fundamentals | R. Sharma |

| Web Development | Yash |

+------------------+-----------+

5 rows in set (0.00 sec)

mysql> SELECT * FROM books;

+---------+------------------+-----------+----------------+------------------+

| book_id | title | author_id | published_year | available_copies |

+---------+------------------+-----------+----------------+------------------+

| 101 | Advanced SQL | 1 | 2018 | 5 |

| 102 | Database Systems | 2 | 2020 | 3 |

| 103 | C Programming | 3 | 2015 | 3 |

| 104 | Java Fundamentals | 1 | 2022 | 6 |

| 105 | Web Development | 2 | 2019 | 2 |

+---------+------------------+-----------+----------------+------------------+

5 rows in set (0.00 sec)

mysql> DELETE FROM books WHERE published_year < 2020;

Query OK, 3 rows affected (0.02 sec)

mysql> SELECT * FROM books;

+---------+------------------+-----------+----------------+------------------+

| book_id | title | author_id | published_year | available_copies |

+---------+------------------+-----------+----------------+------------------+

| 102 | Database Systems | 2 | 2020 | 3 |

| 104 | Java Fundamentals | 1 | 2022 | 6 |

+---------+------------------+-----------+----------------+------------------+

2 rows in set (0.00 sec)

mysql> ELECT * FROM books LEFT JOIN Borrow_Records ON Borrow_Records.book_id IS NULL;

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'ELECT * FROM books LEFT JOIN Borrow_Records ON Borrow_Records.book_id IS NULL' at line 1

mysql> ELECT * FROM books JOIN Borrow_Records ON Borrow_Records.book_id IS NULL;

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'ELECT * FROM books JOIN Borrow_Records ON Borrow_Records.book_id IS NULL' at line 1

mysql> ELECT * FROM books JOIN Borrow_Records WHERE Borrow_Records.book_id IS NULL;

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'ELECT * FROM books JOIN Borrow_Records WHERE Borrow_Records.book_id IS NULL' at line 1

mysql> SELECT * FROM books JOIN Borrow_Records WHERE Borrow_Records.book_id IS NULL;

Empty set (0.00 sec)

mysql> SELECT * FROM books;

+---------+------------------+-----------+----------------+------------------+

| book_id | title | author_id | published_year | available_copies |

+---------+------------------+-----------+----------------+------------------+

| 102 | Database Systems | 2 | 2020 | 3 |

| 104 | Java Fundamentals | 1 | 2022 | 6 |

+---------+------------------+-----------+----------------+------------------+

2 rows in set (0.00 sec)

mysql> SELECT * FROM Borrow_Records;

+-----------+-----------+---------+-------------+-------------+

| record_id | member_id | book_id | borrow_date | return_date |

+-----------+-----------+---------+-------------+-------------+

| 1 | 1 | 101 | 2026-10-01 | 2026-10-10 |

| 2 | 2 | 102 | 2026-10-02 | 2026-10-12 |

| 3 | 3 | 103 | 2026-10-03 | NULL |

| 4 | 1 | 104 | 2026-10-05 | NULL |

+-----------+-----------+---------+-------------+-------------+

4 rows in set (0.00 sec)

mysql> SELECT * FROM books LEFT JOIN Borrow_Recods;

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '' at line 1

mysql> SELECT * FROM books LEFT JOIN Borrow_Recods books.book_id=Borroe_Records.book_id;

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '.book_id=Borroe_Records.book_id' at line 1

```
mysql> SELECT * FROM books LEFT JOIN Borrow_Recods books.book_id=Borroew_Records.book_id;

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '.book_id=Borroew_Records.book_id' at line 1

mysql> SELECT * FROM books LEFT JOIN Borrow_Recods books.book_id=Borrow_Records.book_id;

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '.book_id=Borrow_Records.book_id' at line 1

mysql> SELECT * FROM books LEFT JOIN Borrow_Recods ON books.book_id=Borrow_Records.book_id;

ERROR 1146 (42S02): Table 'library_management_system.borrow_recods' doesn't exist

mysql> SELECT * FROM books LEFT JOIN Borrow_Records ON books.book_id=Borrow_Records.book_id;

+---------+------------------+-----------+----------------+------------------+-----------+-----------+---------+-------------+-------------+

| book_id | title            | author_id | published_year | available_copies | record_id | member_id | book_id | borrow_date | return_date |

+---------+------------------+-----------+----------------+------------------+-----------+-----------+---------+-------------+-------------+

|     102 | Database Systems |         2 |           2020 |                3 |         2 |         2 |     102 | 2026-10-02  | 2026-10-12  |

|     104 | Java Fundamentals |         1 |           2022 |                6 |         4 |         1 |     104 | 2026-10-05  | NULL        |

+---------+------------------+-----------+----------------+------------------+-----------+-----------+---------+-------------+-------------+

2 rows in set (0.00 sec)


mysql> SELECT * FROM books LEFT JOIN Borrow_Records ON books.book_id=Borrow_Records.book_id WHERE Borrow_Records.book_id IS NULL;

Empty set (0.00 sec)


mysql>
```